

## KVITSOY NORDBO, Norway

WMO: 014110

Lat: 59.0705N

Lon: 5.4121E

Elev: 22

StdP: 101.06

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -4.1       | -2.6 | -12.1                         | 1.3 | -2.2 | -9.8 | 1.6 | -1.1 | 18.1                  | 4.2  | 16.3 | 4.2  | 5.0                   | 60   | 0.782 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.5                    | 21.4            | 16.9 | 19.9 | 16.1 | 18.7 | 15.5 | 17.6                | 20.3 | 16.7 | 19.0 | 15.9 | 17.9 | 4.5                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.3                            | 11.7 | 19.0 | 15.6 | 11.1 | 18.1 | 14.9 | 10.6 | 17.3 | 49.5          | 20.2 | 47.0 | 19.2 | 44.8 | 18.0 | 21.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 16.4          | 14.5 | 13.0 | DB                         | -4.3 | 24.8               | 2.3 | 2.2                                                | -6.0 | 26.4       | -7.4 | 27.6       | -8.7  | 28.8       | -10.4 | 30.4 | (4) |
| (5)               |      |      | WB                         | -6.1 | 18.7               | 2.3 | 1.2                                                | -7.7 | 19.5       | -9.1 | 20.2       | -10.3 | 20.8       | -12.0 | 21.7 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 8.8    | 3.2   | 2.7   | 3.7   | 6.7   | 9.8   | 12.6  | 15.3  | 15.5  | 13.6  | 10.2  | 7.0   | 4.8  |
|                                                                                  | DBStd       | 5.15   | 2.72  | 2.66  | 2.43  | 2.28  | 2.78  | 2.23  | 2.39  | 1.87  | 2.07  | 2.25  | 2.72  | 2.99 |
|                                                                                  | HDD10.0     | 1031   | 211   | 205   | 194   | 102   | 37    | 2     | 0     | 0     | 1     | 24    | 94    | 160  |
|                                                                                  | HDD18.3     | 3493   | 469   | 438   | 453   | 349   | 265   | 174   | 101   | 89    | 142   | 253   | 341   | 418  |
|                                                                                  | CDD10.0     | 591    | 0     | 0     | 0     | 4     | 30    | 79    | 163   | 172   | 109   | 30    | 3     | 0    |
|                                                                                  | CDD18.3     | 11     | 0     | 0     | 0     | 0     | 1     | 1     | 6     | 3     | 1     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 14     | 0     | 0     | 0     | 0     | 0     | 1     | 12    | 2     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 6.5    | 7.5   | 6.6   | 6.1   | 5.8   | 6.1   | 6.0   | 5.6   | 5.9   | 6.5   | 6.8   | 7.3   | 7.4  |
| Precipitation                                                                    | PrecAvg     | 1593   | 162   | 130   | 117   | 90    | 76    | 85    | 97    | 144   | 159   | 183   | 175   | 190  |
|                                                                                  | PrecMax     | 1928   | 297   | 248   | 230   | 155   | 149   | 154   | 165   | 258   | 269   | 283   | 320   | 315  |
|                                                                                  | PrecMin     | 1069   | 11    | 28    | 2     | 41    | 25    | 24    | 46    | 36    | 45    | 61    | 35    | 39   |
|                                                                                  | PrecStd     | 230    | 75    | 65    | 55    | 26    | 30    | 35    | 31    | 48    | 67    | 65    | 76    | 72   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 9.0   | 8.0   | 10.4  | 15.3  | 20.5  | 21.9  | 25.0  | 22.5  | 20.6  | 16.5  | 13.0  | 10.5 |
|                                                                                  |             | MCWB   | 8.0   | 6.9   | 7.0   | 10.3  | 15.2  | 15.8  | 18.3  | 17.1  | 16.2  | 13.3  | 11.0  | 9.4  |
|                                                                                  | 2%          | DB     | 8.1   | 7.2   | 8.0   | 13.1  | 17.6  | 19.2  | 22.4  | 20.5  | 18.5  | 14.9  | 12.0  | 9.7  |
|                                                                                  |             | MCWB   | 7.0   | 6.2   | 5.8   | 9.4   | 12.8  | 14.9  | 17.8  | 16.5  | 15.1  | 12.6  | 10.6  | 8.5  |
|                                                                                  | 5%          | DB     | 7.5   | 6.7   | 7.0   | 11.4  | 15.8  | 17.1  | 20.2  | 19.4  | 17.3  | 13.8  | 11.2  | 9.1  |
|                                                                                  |             | MCWB   | 6.4   | 5.7   | 5.5   | 8.5   | 11.9  | 14.1  | 16.7  | 16.0  | 14.6  | 11.8  | 9.7   | 8.0  |
|                                                                                  | 10%         | DB     | 7.0   | 6.3   | 6.5   | 9.8   | 13.8  | 15.7  | 18.7  | 18.3  | 16.4  | 13.1  | 10.5  | 8.5  |
|                                                                                  |             | MCWB   | 5.8   | 5.3   | 5.2   | 7.8   | 10.8  | 13.1  | 16.0  | 15.5  | 14.1  | 11.2  | 8.9   | 7.3  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 8.2   | 7.4   | 7.4   | 10.7  | 16.0  | 16.6  | 19.4  | 18.2  | 17.0  | 13.9  | 11.7  | 9.7  |
|                                                                                  |             | MCDB   | 8.6   | 7.7   | 9.2   | 14.0  | 20.1  | 19.8  | 23.4  | 20.7  | 19.2  | 15.3  | 12.4  | 10.2 |
|                                                                                  | 2%          | WB     | 7.3   | 6.5   | 6.4   | 9.5   | 13.8  | 15.5  | 18.1  | 17.2  | 15.9  | 13.1  | 10.9  | 8.9  |
|                                                                                  |             | MCDB   | 7.9   | 7.0   | 7.5   | 11.5  | 17.0  | 18.5  | 21.3  | 19.4  | 17.4  | 14.4  | 11.8  | 9.5  |
|                                                                                  | 5%          | WB     | 6.7   | 6.0   | 5.9   | 8.7   | 12.4  | 14.4  | 17.2  | 16.4  | 15.2  | 12.4  | 10.1  | 8.2  |
|                                                                                  |             | MCDB   | 7.4   | 6.6   | 6.6   | 10.5  | 15.3  | 16.8  | 19.7  | 18.5  | 17.0  | 13.6  | 11.0  | 9.0  |
|                                                                                  | 10%         | WB     | 6.0   | 5.4   | 5.4   | 7.8   | 11.3  | 13.4  | 16.2  | 15.8  | 14.4  | 11.6  | 9.3   | 7.5  |
|                                                                                  |             | MCDB   | 6.8   | 6.1   | 6.2   | 9.1   | 13.6  | 15.3  | 18.3  | 17.7  | 16.1  | 12.9  | 10.3  | 8.4  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 2.7   | 2.6   | 3.1   | 3.9   | 4.1   | 3.9   | 3.8   | 3.5   | 3.1   | 2.9   | 2.8   | 2.7  |
|                                                                                  |             | MCDBR  | 3.4   | 2.7   | 3.4   | 6.1   | 7.3   | 6.3   | 6.3   | 4.9   | 4.0   | 3.3   | 3.0   | 3.1  |
|                                                                                  |             | MCWBR  | 3.7   | 2.7   | 2.5   | 3.4   | 3.9   | 3.1   | 3.2   | 2.9   | 2.8   | 2.8   | 3.2   | 3.4  |
|                                                                                  | 5% WB       | MCDBR  | 3.2   | 2.7   | 3.1   | 5.1   | 6.9   | 5.6   | 5.7   | 4.2   | 3.6   | 3.2   | 3.1   | 3.1  |
|                                                                                  |             | MCWBR  | 3.7   | 2.8   | 2.6   | 3.2   | 4.0   | 3.3   | 3.1   | 3.1   | 3.1   | 3.2   | 3.6   | 3.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.308  | 0.331 | 0.351 | 0.381 | 0.368 | 0.363 | 0.377 | 0.377 | 0.359 | 0.339 | 0.311 | 0.285 |      |
|                                                                                  | taud        | 2.179  | 2.287 | 2.337 | 2.307 | 2.419 | 2.465 | 2.445 | 2.442 | 2.477 | 2.416 | 2.253 | 2.120 |      |
|                                                                                  | Ebn at Noon | 481    | 678   | 785   | 821   | 863   | 873   | 849   | 821   | 775   | 669   | 479   | 366   |      |
|                                                                                  | Edh at Noon | 47     | 70    | 89    | 108   | 104   | 101   | 101   | 94    | 78    | 62    | 44    | 35    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.28   | 0.87  | 2.07  | 3.79  | 5.20  | 5.57  | 4.92  | 3.79  | 2.43  | 1.17  | 0.40  | 0.18  |      |
|                                                                                  | RadStd      | 0.04   | 0.14  | 0.24  | 0.34  | 0.55  | 0.64  | 0.47  | 0.33  | 0.21  | 0.14  | 0.05  | 0.02  |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (32 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -116    | N/A     | N/A     |

Nomenclature: See separate page